Analytics Vidhya

Introduction to Load Testing

Instructor

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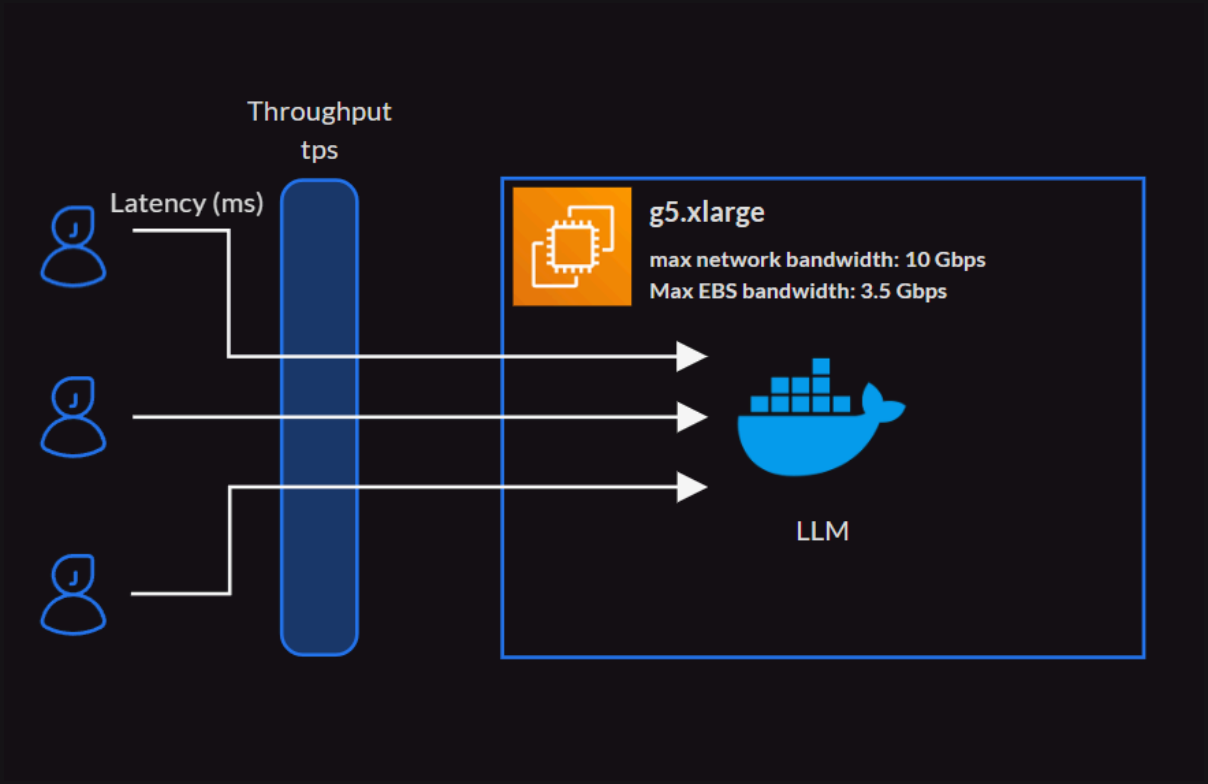
Certified AWS Cloud and Kubernetes Engineer

Open source contributor

LLM Inference

- Latency vs throughput vs bandwidth
- Throughput depends on the number of concurrent users and is measured in tokens per second (tps)
- Higher the concurrent users higher the latency



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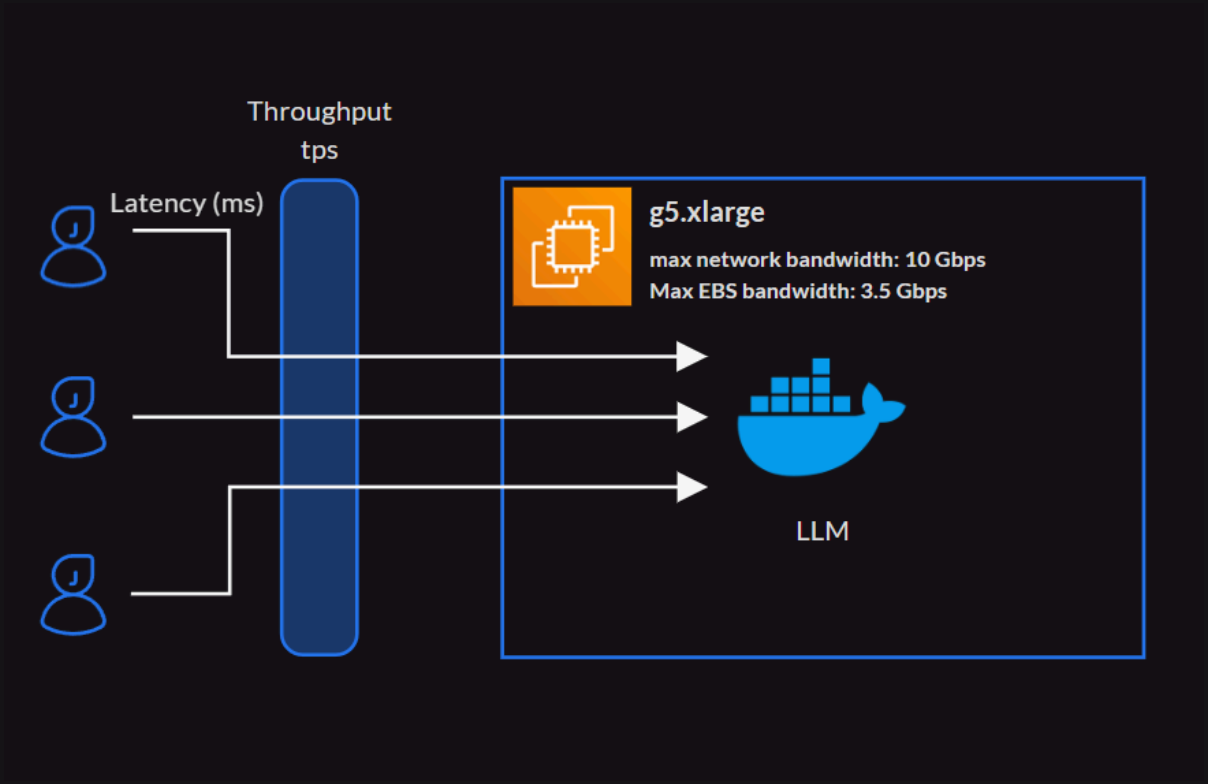
LLM Inference

LLM inference (TTFT vs TPS)

- Prefill phase:
  - Compute heavy
  - Small delay we see in UI
- Decode phase:
  - Memory transfer overhead
  - Higher batch size is better

- How to measure this for a given set of users ?



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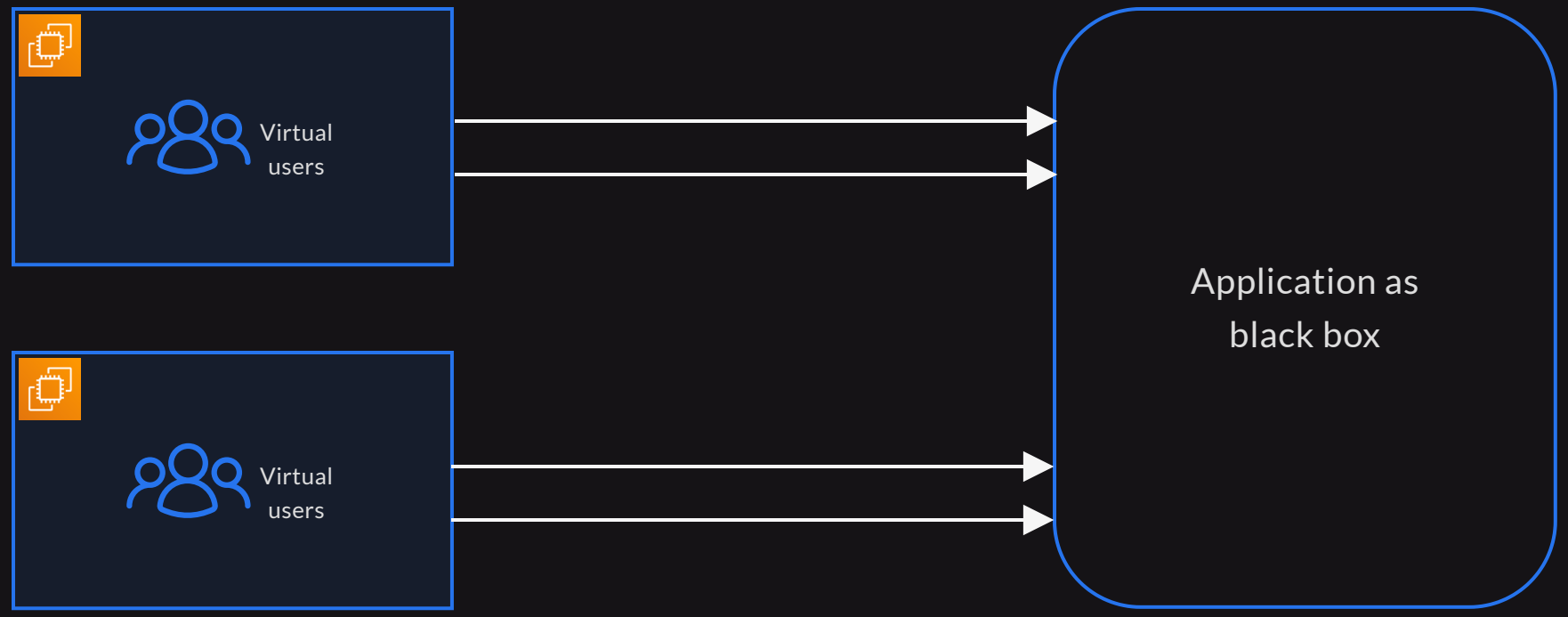
Load Testing



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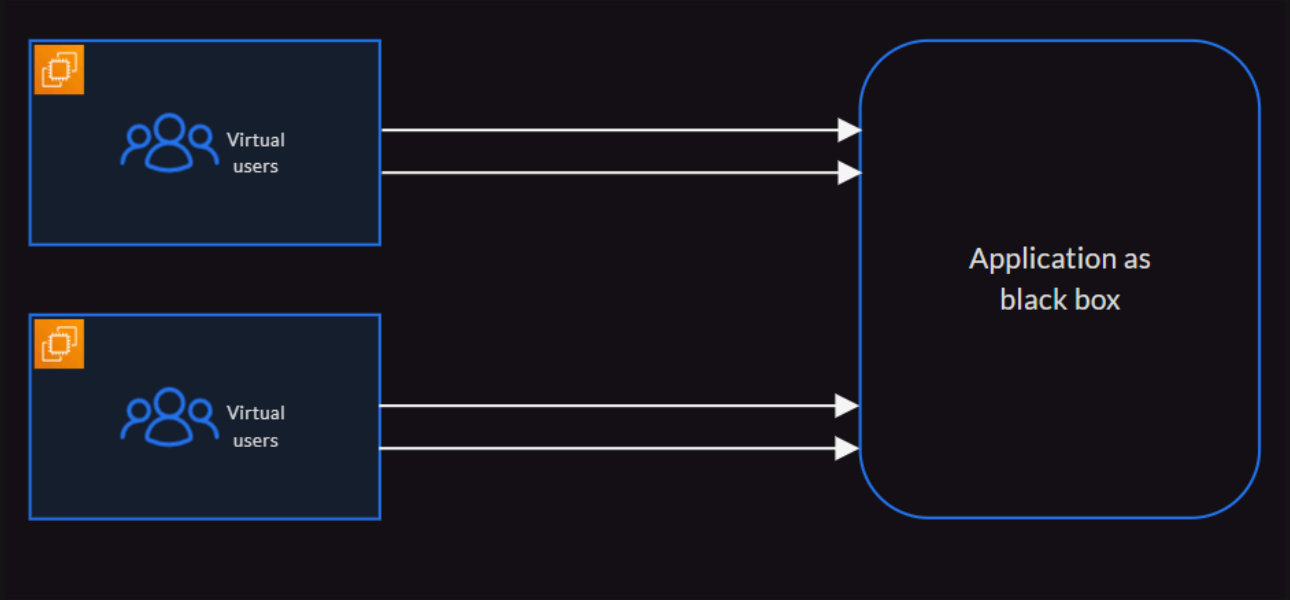
Load Testing

- Black box test focuses on testing the system's behavior and features and not how it works internally
- Load testing is a subset of performance testing that generally looks at how a system responds to normal and peak usage
- We are looking for slow response times, errors, crashes, and other issues to determine how many users and transactions the system can accommodate before performance suffers
- We simulate real-user behaviour using code also known as virtual users
- These requests are then bombarded to the application and critical metrics are measured



Thank You